
STM32
CubeMX

# 1. Description

## 1.1. Project

| | |
|---|---|
| Project Name | yuancheng |
| Board Name | custom |
| Generated with: | STM32CubeMX 6.18.1 |
| Date | 10/19/2026 |

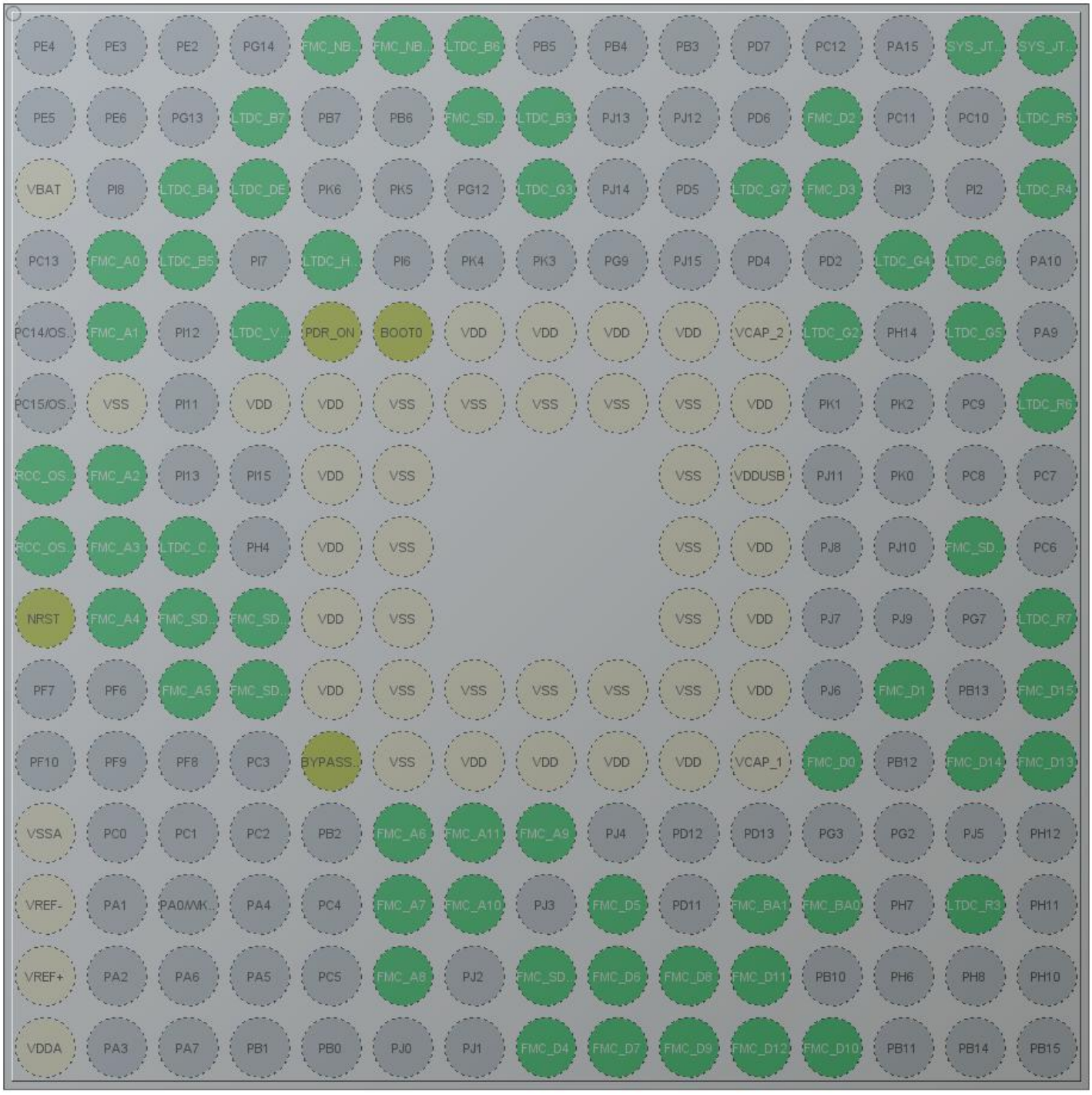
## 1.2. MCU

| | |
|---|---|
| MCU Series | STM32F7 |
| MCU Line | STM32F7x6 |
| MCU name | STM32F746NGHx |
| MCU Package | TFBGA216 |
| MCU Pin number | 216 |

## 1.3. Core(s) information

| | |
|---|---|
| Core(s) | Arm Cortex-M7 |

## 2. Pinout Configuration




**TFBGA216 (Top view)**

# 3. Pins Configuration

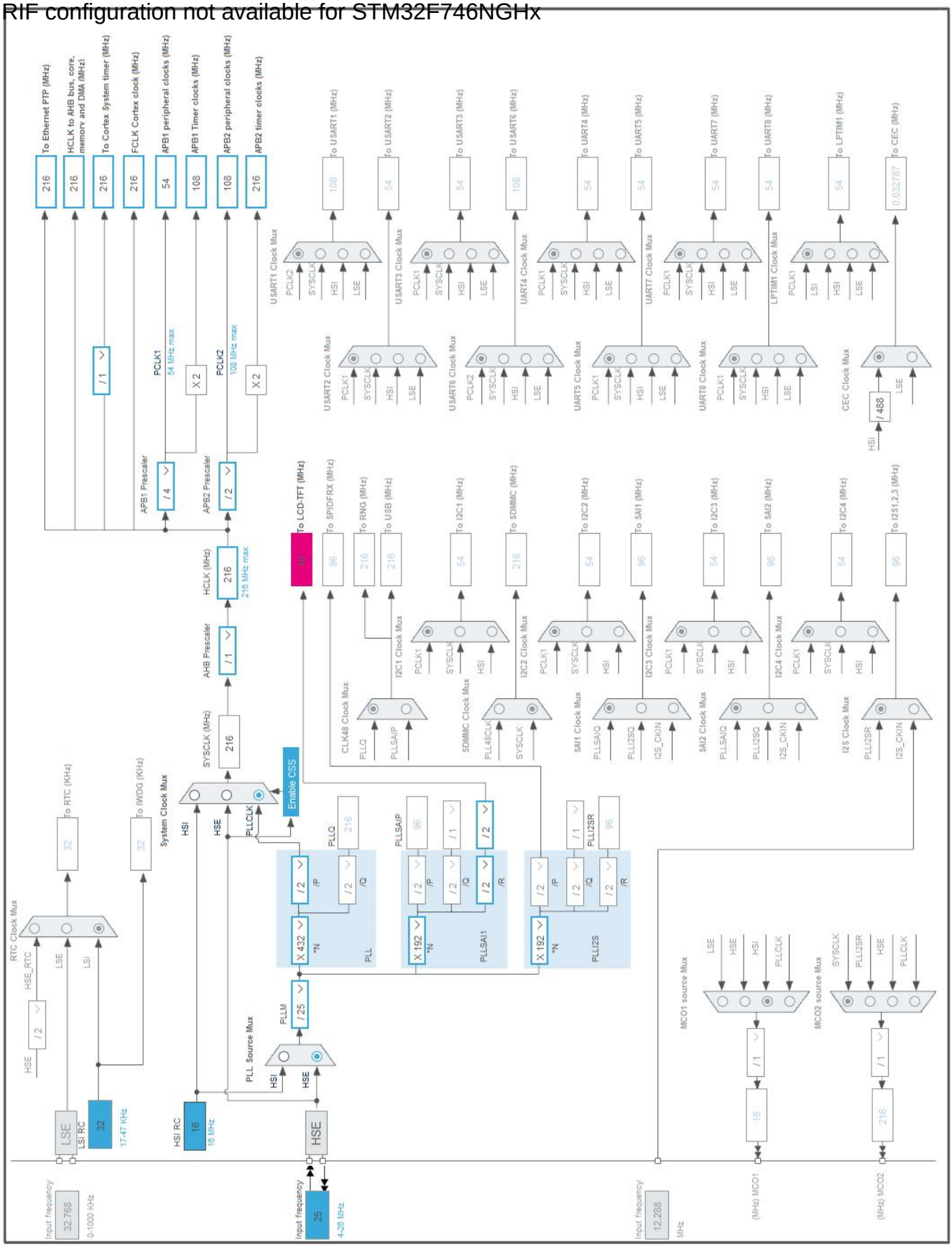
| Pin Number TFBGA216 | Pin Name (function after reset) | Pin Type | Alternate Function(s) | Label |
|---|---|---|---|---|
| A5 | PE1 | I/O | FMC_NBL1 | |
| A6 | PE0 | I/O | FMC_NBL0 | |
| A7 | PB8 | I/O | LTDC_B6 | |
| A14 | PA14 | I/O | SYS_JTCK-SWCLK | |
| A15 | PA13 | I/O | SYS_JTMS-SWDIO | |
| B4 | PB9 | I/O | LTDC_B7 | |
| B7 | PG15 | I/O | FMC_SDNCAS | |
| B8 | PG11 | I/O | LTDC_B3 | |
| B12 | PD0 | I/O | FMC_D2 | |
| B15 | PA12 | I/O | LTDC_R5 | |
| C1 | VBAT | Power | | |
| C3 | PI4 | I/O | LTDC_B4 | |
| C4 | PK7 | I/O | LTDC_DE | |
| C8 | PG10 | I/O | LTDC_G3 | |
| C11 | PD3 | I/O | LTDC_G7 | |
| C12 | PD1 | I/O | FMC_D3 | |
| C15 | PA11 | I/O | LTDC_R4 | |
| D2 | PF0 | I/O | FMC_A0 | |
| D3 | PI5 | I/O | LTDC_B5 | |
| D5 | PI10 | I/O | LTDC_HSYNC | |
| D13 | PH15 | I/O | LTDC_G4 | |
| D14 | PI1 | I/O | LTDC_G6 | |
| E2 | PF1 | I/O | FMC_A1 | |
| E4 | PI9 | I/O | LTDC_VSYNC | |
| E5 | PDR_ON | Reset | | |
| E6 | BOOT0 | Boot | | |
| E7 | VDD | Power | | |
| E8 | VDD | Power | | |
| E9 | VDD | Power | | |
| E10 | VDD | Power | | |
| E11 | VCAP_2 | Power | | |
| E12 | PH13 | I/O | LTDC_G2 | |
| E14 | PI0 | I/O | LTDC_G5 | |
| F2 | VSS | Power | | |
| F4 | VDD | Power | | |
| F5 | VDD | Power | | |

| Pin Number TFBGA216 | Pin Name (function after reset) | Pin Type | Alternate Function(s) | Label |
|---|---|---|---|---|
| F6 | VSS | Power | | |
| F7 | VSS | Power | | |
| F8 | VSS | Power | | |
| F9 | VSS | Power | | |
| F10 | VSS | Power | | |
| F11 | VDD | Power | | |
| F15 | PA8 | I/O | LTDC_R6 | |
| G1 | PH0/OSC_IN | I/O | RCC_OSC_IN | |
| G2 | PF2 | I/O | FMC_A2 | |
| G5 | VDD | Power | | |
| G6 | VSS | Power | | |
| G10 | VSS | Power | | |
| G11 | VDDUSB | Power | | |
| H1 | PH1/OSC_OUT | I/O | RCC_OSC_OUT | |
| H2 | PF3 | I/O | FMC_A3 | |
| H3 | PI14 | I/O | LTDC_CLK | |
| H5 | VDD | Power | | |
| H6 | VSS | Power | | |
| H10 | VSS | Power | | |
| H11 | VDD | Power | | |
| H14 | PG8 | I/O | FMC_SDCLK | |
| J1 | NRST | Reset | | |
| J2 | PF4 | I/O | FMC_A4 | |
| J3 | PH5 | I/O | FMC_SDNWE | |
| J4 | PH3 | I/O | FMC_SDNE0 | |
| J5 | VDD | Power | | |
| J6 | VSS | Power | | |
| J10 | VSS | Power | | |
| J11 | VDD | Power | | |
| J15 | PG6 | I/O | LTDC_R7 | |
| K3 | PF5 | I/O | FMC_A5 | |
| K4 | PH2 | I/O | FMC_SDCKE0 | |
| K5 | VDD | Power | | |
| K6 | VSS | Power | | |
| K7 | VSS | Power | | |
| K8 | VSS | Power | | |
| K9 | VSS | Power | | |
| K10 | VSS | Power | | |
| K11 | VDD | Power | | |

| Pin Number TFBGA216 | Pin Name (function after reset) | Pin Type | Alternate Function(s) | Label |
|---|---|---|---|---|
| K13 | PD15 | I/O | FMC_D1 | |
| K15 | PD10 | I/O | FMC_D15 | |
| L5 | BYPASS_REG | Reset | | |
| L6 | VSS | Power | | |
| L7 | VDD | Power | | |
| L8 | VDD | Power | | |
| L9 | VDD | Power | | |
| L10 | VDD | Power | | |
| L11 | VCAP_1 | Power | | |
| L12 | PD14 | I/O | FMC_D0 | |
| L14 | PD9 | I/O | FMC_D14 | |
| L15 | PD8 | I/O | FMC_D13 | |
| M1 | VSSA | Power | | |
| M6 | PF12 | I/O | FMC_A6 | |
| M7 | PG1 | I/O | FMC_A11 | |
| M8 | PF15 | I/O | FMC_A9 | |
| N1 | VREF- | Power | | |
| N6 | PF13 | I/O | FMC_A7 | |
| N7 | PG0 | I/O | FMC_A10 | |
| N9 | PE8 | I/O | FMC_D5 | |
| N11 | PG5 | I/O | FMC_BA1 | |
| N12 | PG4 | I/O | FMC_BA0 | |
| N14 | PH9 | I/O | LTDC_R3 | |
| P1 | VREF+ | Power | | |
| P6 | PF14 | I/O | FMC_A8 | |
| P8 | PF11 | I/O | FMC_SDNRAS | |
| P9 | PE9 | I/O | FMC_D6 | |
| P10 | PE11 | I/O | FMC_D8 | |
| P11 | PE14 | I/O | FMC_D11 | |
| R1 | VDDA | Power | | |
| R8 | PE7 | I/O | FMC_D4 | |
| R9 | PE10 | I/O | FMC_D7 | |
| R10 | PE12 | I/O | FMC_D9 | |
| R11 | PE15 | I/O | FMC_D12 | |
| R12 | PE13 | I/O | FMC_D10 | |

# 4. Clock Tree Configuration

RIF configuration not available for STM32F746NGHx



# 1. Power Consumption Calculator report

## 1.1. Microcontroller Selection

| | |
|---|---|
| Series | STM32F7 |
| Line | STM32F7x6 |
| MCU | STM32F746NGHx |
| Datasheet | DS10916_Rev4 |

## 1.2. Parameter Selection

| | |
|---|---|
| Temperature | 25 |
| Vdd | 3.3 |

## 1.3. Battery Selection

| | |
|---|---|
| Battery | Alkaline(9V) |
| Capacity | 625.0 mAh |
| Self Discharge | 0.3 %/month |
| Nominal Voltage | 9.0 V |
| Max Cont Current | 200.0 mA |
| Max Pulse Current | 0.0 mA |
| Cells in series | 1 |
| Cells in parallel | 1 |

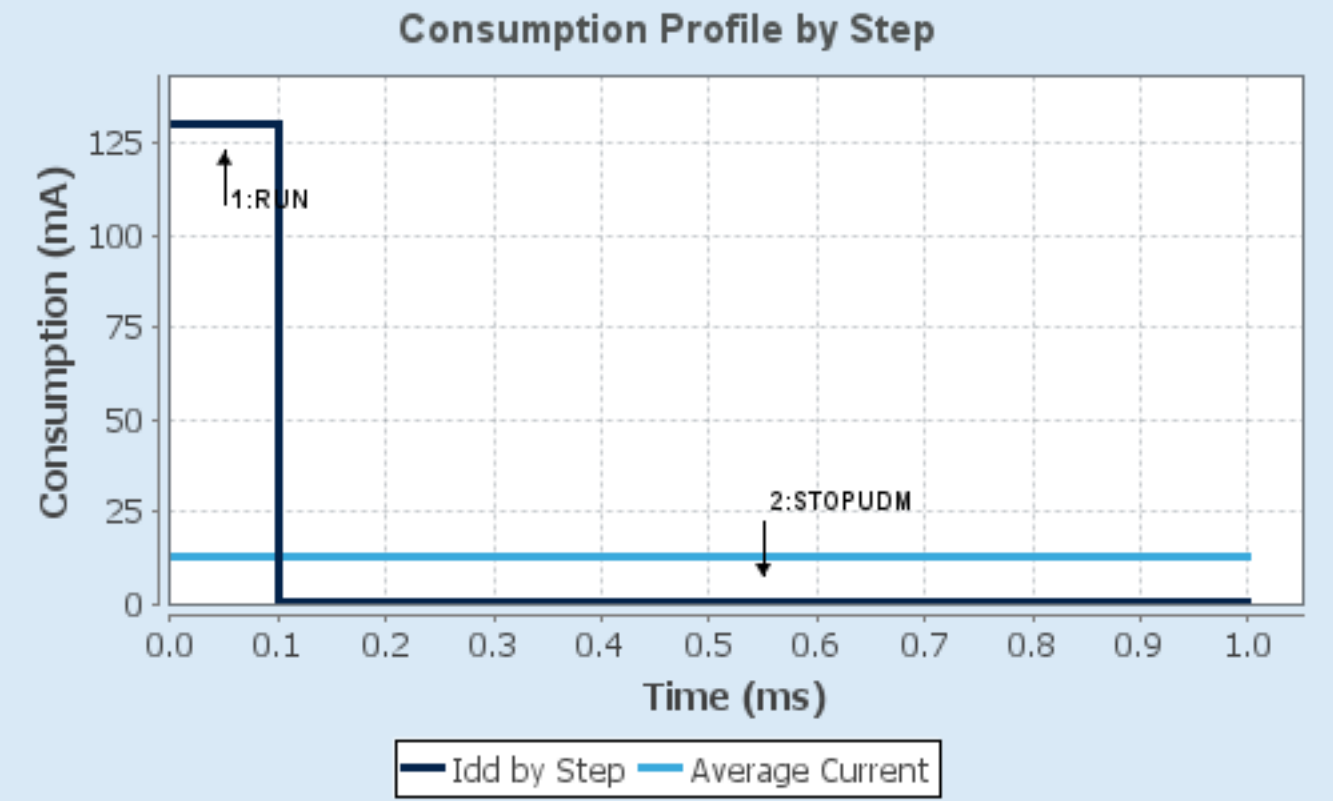
## 1.4. Sequence

| ***Step*** | Step1 | Step2 |
|---|---|---|
| ***Mode*** | RUN | STOP_UDM (Under Drive) |
| ***Vdd*** | 3.3 | 3.3 |
| ***Voltage Source*** | Battery | Battery |
| ***Range*** | Scale1-High | No Scale |
| ***Fetch Type*** | ITCM/FLASH/REGON | n/a |
| ***CPU Frequency*** | 216 MHz | 0 Hz |
| ***Clock Configuration*** | HSE PLL | Regulator_LP Flash-PwrDwn |
| ***Clock Source Frequency*** | 4 MHz | 0 Hz |
| ***Peripherals*** | | |
| ***Additional Cons.*** | 0 mA | 0 mA |
| ***Average Current*** | 130 mA | 100 µA |
| ***Duration*** | 0.1 ms | 0.9 ms |
| ***DMIPS*** | 462.0 | 0.0 |
| ***Ta Max*** | 92.56 | 104.99 |
| ***Category*** | In DS Table | In DS Table |

## 1.5. Results

| Sequence Time | 1 ms | Average Current | 13.09 mA |
|---|---|---|---|
| Battery Life | 1 day, 23 hours | Average DMIPS | 462.24005 DMIPS |

## 1.6. Chart


Consumption Profile by Step
Consumption (mA)
125
100
75
50
25
0
1:RUN
2:STOPUDM
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Time (ms)
Idd by Step
Average Current

# 2. Software Project

## 2.1. Project Settings

| Name | Value |
|---|---|
| Project Name | yuancheng |
| Project Folder | D:\stm32_Project\qianrushi\yuancheng |
| Toolchain / IDE | EWARM V8.50 |
| Firmware Package Name and Version | STM32Cube FW_F7 V1.17.4 |
| Application Structure | Advanced |
| Generate Under Root | No |
| Do not generate the main() | No |
| Minimum Heap Size | 0x200 |
| Minimum Stack Size | 0x400 |

## 2.2. Code Generation Settings

| Name | Value |
|---|---|
| STM32Cube MCU packages and embedded software | Copy all used libraries into the project folder |
| Generate peripheral initialization as a pair of '.c/.h' files | Yes |
| Backup previously generated files when re-generating | No |
| Keep User Code when re-generating | Yes |
| Delete previously generated files when not re-generated | Yes |
| Set all free pins as analog (to optimize the power consumption) | No |
| Enable Full Assert | No |

## 2.3. Advanced Settings - Generated Function Calls

| Rank | Function Name | Peripheral Instance Name |
|---|---|---|
| 1 | SystemClock_Config | RCC |
| 2 | MX_GPIO_Init | GPIO |
| 3 | MX_FMC_Init | FMC |
| 4 | MX_LTDC_Init | LTDC |

# 3. Peripherals and Middlewares Configuration

## 3.1. CORTEX_M7

### 3.1.1. Parameter Settings:

**Speculation default mode Settings:**

| | |
|---|---|
| Speculation default mode | **Enabled *** |

**Cortex Interface Settings:**

| | |
|---|---|
| Flash Interface | AXI Interface |
| ART ACCELERATOR | Disabled |
| Instruction Prefetch | Disabled |
| CPU ICache | Disabled |
| CPU DCache | Disabled |

**Cortex Memory Protection Unit Control Settings:**

| | |
|---|---|
| MPU Control Mode | Background Region Privileged accesses only + MPU Disabled during hard fault, NMI and FAULTMASK handlers |

**Cortex Memory Protection Unit Region 0 Settings:**

| | |
|---|---|
| MPU Region | Enabled |
| MPU Region Base Address | **0x0 *** |
| MPU Region Size | 4GB |
| MPU SubRegion Disable | **0x87 *** |
| MPU TEX field level | level 0 |
| MPU Access Permission | ALL ACCESS NOT PERMITTED |
| MPU Instruction Access | DISABLE |
| MPU Shareability Permission | ENABLE |
| MPU Cacheable Permission | DISABLE |
| MPU Bufferable  Permission | DISABLE |

**Cortex Memory Protection Unit Region 1 Settings:**

| | |
|---|---|
| MPU Region | Disabled |

**Cortex Memory Protection Unit Region 2 Settings:**

| | |
|---|---|
| MPU Region | Disabled |

**Cortex Memory Protection Unit Region 3 Settings:**

| | |
|---|---|
| MPU Region | Disabled |

**Cortex Memory Protection Unit Region 4 Settings:**

| | |
|---|---|
| MPU Region | Disabled |

**Cortex Memory Protection Unit Region 5 Settings:**

| | |
|---|---|
| MPU Region | Disabled |

**Cortex Memory Protection Unit Region 6 Settings:**

| | |
|---|---|
| MPU Region | Disabled |

**Cortex Memory Protection Unit Region 7 Settings:**

| | |
|---|---|
| MPU Region | Disabled |

## 3.2. FMC

*SDRAM 1*

**Clock and chip enable: SDCKE0+SDNE0**

**Internal bank number: 4 banks**

**Address: 12 bits**

**Data: 16 bits**

**Byte enable: 16-bit byte enable**

### 3.2.1. SDRAM 1:

**SDRAM control:**

| | |
|---|---|
| Bank | SDRAM bank 1 |
| Number of column address bits | 8 bits |
| Number of row address bits | 12 bits |
| CAS latency | **3 memory clock cycles *** |
| Write protection | Disabled |
| SDRAM common clock | **2 HCLK clock cycles *** |
| SDRAM common burst read | **Enabled *** |
| SDRAM common read pipe delay | **1 HCLK clock cycle *** |

**SDRAM timing in memory clock cycles:**

| | |
|---|---|
| Load mode register to active delay | **2 *** |
| Exit self-refresh delay | **7 *** |
| Self-refresh time | **7 *** |
| SDRAM common row cycle delay | **4 *** |
| Write recovery time | 16 |
| SDRAM common row precharge delay | **2 *** |
| Row to column delay | **2 *** |

## 3.3. LTDC

**Display Type: RGB565 (16 bits)**

### 3.3.1. Parameter Settings:

**Synchronization for Width:**

| | |
|---|---|
| Horizontal Synchronization Width | 8 |

| | |
|---|---|
| Horizontal Back Porch | 7 |
| Active Width | 640 |
| Horizontal Front Porch | 6 |
| HSync Width | 7 |
| Accumulated Horizontal Back Porch Width | 14 |
| Accumulated Active Width | 654 |
| Total Width | 660 |

**Synchronization for Height:**

| | |
|---|---|
| Vertical Synchronization Height | 4 |
| Vertical Back Porch | 2 |
| Active Height | 480 |
| Vertical Front Porch | 2 |
| VSync Height | 3 |
| Accumulated Vertical Back Porch Height | 5 |
| Accumulated Active Height | 485 |
| Total Height | 487 |

**Signal Polarity:**

| | |
|---|---|
| Horizontal Synchronization Polarity | Active Low |
| Vertical Synchronization Polarity | Active Low |
| Data Enable Polarity | Active Low |
| Pixel Clock Polarity | Normal Input |

**Layer Default Color:**

| | |
|---|---|
| Red | 0 |
| Green | 0 |
| Blue | 0 |

## 3.3.2. Layer Settings:

**Layer Default Color:**

| | |
|---|---|
| Layer 0 - Alpha | 0 |
| Layer 0 - Blue | 0 |
| Layer 0 - Green | 0 |
| Layer 0 - Red | 0 |
| Layer 1 - Alpha | 0 |
| Layer 1 - Blue | 0 |
| Layer 1 - Green | 0 |
| Layer 1 - Red | 0 |

**Windows Position:**

| | |
|---|---|
| Layer 0 - Window Horizontal Start | 0 |
| Layer 0 - Window Horizontal Stop | 0 |
| Layer 0 - Window Vertical Start | 0 |

| | |
|---|---|
| Layer 0 - Window Vertical Stop | 0 |
| Layer 1 - Window Horizontal Start | 0 |
| Layer 1 - Window Horizontal Stop | 0 |
| Layer 1 - Window Vertical Start | 0 |
| Layer 1 - Window Vertical Stop | 0 |

**Pixel Parameters:**

| | |
|---|---|
| Layer 0 - Pixel Format | ARGB8888 |
| Layer 1 - Pixel Format | ARGB8888 |

**Blending:**

| | |
|---|---|
| Layer 0 - Alpha constant for blending | 0 |
| Layer 0 - Blending Factor1 | Alpha constant |
| Layer 0 - Blending Factor2 | Alpha constant |
| Layer 1 - Alpha constant for blending | 0 |
| Layer 1 - Blending Factor1 | Alpha constant |
| Layer 1 - Blending Factor2 | Alpha constant |

**Frame Buffer:**

| | |
|---|---|
| Layer 0 - Color Frame Buffer Start Adress | 0 |
| Layer 0 - Color Frame Buffer Line Length (Image Width) | 0 |
| Layer 0 - Color Frame Buffer Number of Lines (Image Height) | 0 |
| Layer 1 - Color Frame Buffer Start Adress | 0 |
| Layer 1 - Color Frame Buffer Line Length (Image Width) | 0 |
| Layer 1 - Color Frame Buffer Number of Lines (Image Height) | 0 |

**Number of Layers:**

| | |
|---|---|
| Number of Layers | 2 layers |

## 3.4. RCC

## High Speed Clock (HSE): Crystal/Ceramic Resonator

### 3.4.1. Parameter Settings:

**System Parameters:**

| | |
|---|---|
| VDD voltage (V) | 3.3 |
| Flash Latency(WS) | 7 WS (8 CPU cycle) |

**RCC Parameters:**

| | |
|---|---|
| HSI Calibration Value | 16 |
| TIM Prescaler Selection | Disabled |
| HSE Startup Timout Value (ms) | 100 |

| | |
|---|---|
| LSE Startup Timout Value (ms) | 5000 |

**Power Parameters:**

| | |
|---|---|
| Power Over Drive | Enabled |
| Power Regulator Voltage Scale | Power Regulator Voltage Scale 1 |

## 3.5. SYS

**Debug: Serial Wire**

**Timebase Source: SysTick**

*** User modified value**

# 4. System Configuration

## 4.1. GPIO configuration

| IP | Pin | Signal | GPIO mode | GPIO pull/up pull down | Max Speed | User Label |
|---|---|---|---|---|---|---|
| FMC | PE1 | FMC_NBL1 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PE0 | FMC_NBL0 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PG15 | FMC_SDNCAS | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PD0 | FMC_D2 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PD1 | FMC_D3 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PF0 | FMC_A0 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PF1 | FMC_A1 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PF2 | FMC_A2 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PF3 | FMC_A3 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PG8 | FMC_SDCLK | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PF4 | FMC_A4 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PH5 | FMC_SDNWE | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PH3 | FMC_SDNE0 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PF5 | FMC_A5 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PH2 | FMC_SDCKE0 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PD15 | FMC_D1 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PD10 | FMC_D15 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PD14 | FMC_D0 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PD9 | FMC_D14 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PD8 | FMC_D13 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PF12 | FMC_A6 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PG1 | FMC_A11 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PF15 | FMC_A9 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PF13 | FMC_A7 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PG0 | FMC_A10 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PE8 | FMC_D5 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PG5 | FMC_BA1 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PG4 | FMC_BA0 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PF14 | FMC_A8 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PF11 | FMC_SDNRAS | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PE9 | FMC_D6 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PE11 | FMC_D8 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PE14 | FMC_D11 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PE7 | FMC_D4 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PE10 | FMC_D7 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PE12 | FMC_D9 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | | | | | | |

| IP | Pin | Signal | GPIO mode | GPIO pull/up pull down | Max Speed | User Label |
|---|---|---|---|---|---|---|
| | PE15 | FMC_D12 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| | PE13 | FMC_D10 | Alternate Function Push Pull | No pull-up and no pull-down | Very High | |
| LTDC | PB8 | LTDC_B6 | Alternate Function Push Pull | No pull-up and no pull-down | Low | |
| | PB9 | LTDC_B7 | Alternate Function Push Pull | No pull-up and no pull-down | Low | |
| | PG11 | LTDC_B3 | Alternate Function Push Pull | No pull-up and no pull-down | Low | |
| | PA12 | LTDC_R5 | Alternate Function Push Pull | No pull-up and no pull-down | Low | |
| | PI4 | LTDC_B4 | Alternate Function Push Pull | No pull-up and no pull-down | Low | |
| | PK7 | LTDC_DE | Alternate Function Push Pull | No pull-up and no pull-down | Low | |
| | PG10 | LTDC_G3 | Alternate Function Push Pull | No pull-up and no pull-down | Low | |
| | PD3 | LTDC_G7 | Alternate Function Push Pull | No pull-up and no pull-down | Low | |
| | PA11 | LTDC_R4 | Alternate Function Push Pull | No pull-up and no pull-down | Low | |
| | PI5 | LTDC_B5 | Alternate Function Push Pull | No pull-up and no pull-down | Low | |
| | PI10 | LTDC_HSYNC | Alternate Function Push Pull | No pull-up and no pull-down | Low | |
| | PH15 | LTDC_G4 | Alternate Function Push Pull | No pull-up and no pull-down | Low | |
| | PI1 | LTDC_G6 | Alternate Function Push Pull | No pull-up and no pull-down | Low | |
| | PI9 | LTDC_VSYNC | Alternate Function Push Pull | No pull-up and no pull-down | Low | |
| | PH13 | LTDC_G2 | Alternate Function Push Pull | No pull-up and no pull-down | Low | |
| | PI0 | LTDC_G5 | Alternate Function Push Pull | No pull-up and no pull-down | Low | |
| | PA8 | LTDC_R6 | Alternate Function Push Pull | No pull-up and no pull-down | Low | |
| | PI14 | LTDC_CLK | Alternate Function Push Pull | No pull-up and no pull-down | Low | |
| | PG6 | LTDC_R7 | Alternate Function Push Pull | No pull-up and no pull-down | Low | |
| | PH9 | LTDC_R3 | Alternate Function Push Pull | No pull-up and no pull-down | Low | |
| RCC | PH0/OSC_IN | RCC_OSC_IN | n/a | n/a | n/a | |
| | PH1/OSC_OUT | RCC_OSC_OUT | n/a | n/a | n/a | |
| SYS | PA14 | SYS_JTCK-SWCLK | n/a | n/a | n/a | |
| | PA13 | SYS_JTMS-SWDIO | n/a | n/a | n/a | |

### 4.2. DMA configuration

nothing configured in DMA service

## 4.3. NVIC configuration

### 4.3.1. NVIC

| Interrupt Table | Enable | Preenmption Priority | SubPriority |
|---|---|---|---|
| Non maskable interrupt | true | 0 | 0 |
| Hard fault interrupt | true | 0 | 0 |
| Memory management fault | true | 0 | 0 |
| Pre-fetch fault, memory access fault | true | 0 | 0 |
| Undefined instruction or illegal state | true | 0 | 0 |
| System service call via SWI instruction | true | 0 | 0 |
| Debug monitor | true | 0 | 0 |
| Pendable request for system service | true | 0 | 0 |
| System tick timer | true | 15 | 0 |
| PVD interrupt through EXTI line 16 | unused | | |
| Flash global interrupt | unused | | |
| RCC global interrupt | unused | | |
| FMC global interrupt | unused | | |
| FPU global interrupt | unused | | |
| LTDC global interrupt | unused | | |
| LTDC global error interrupt | unused | | |

### 4.3.2. NVIC Code generation

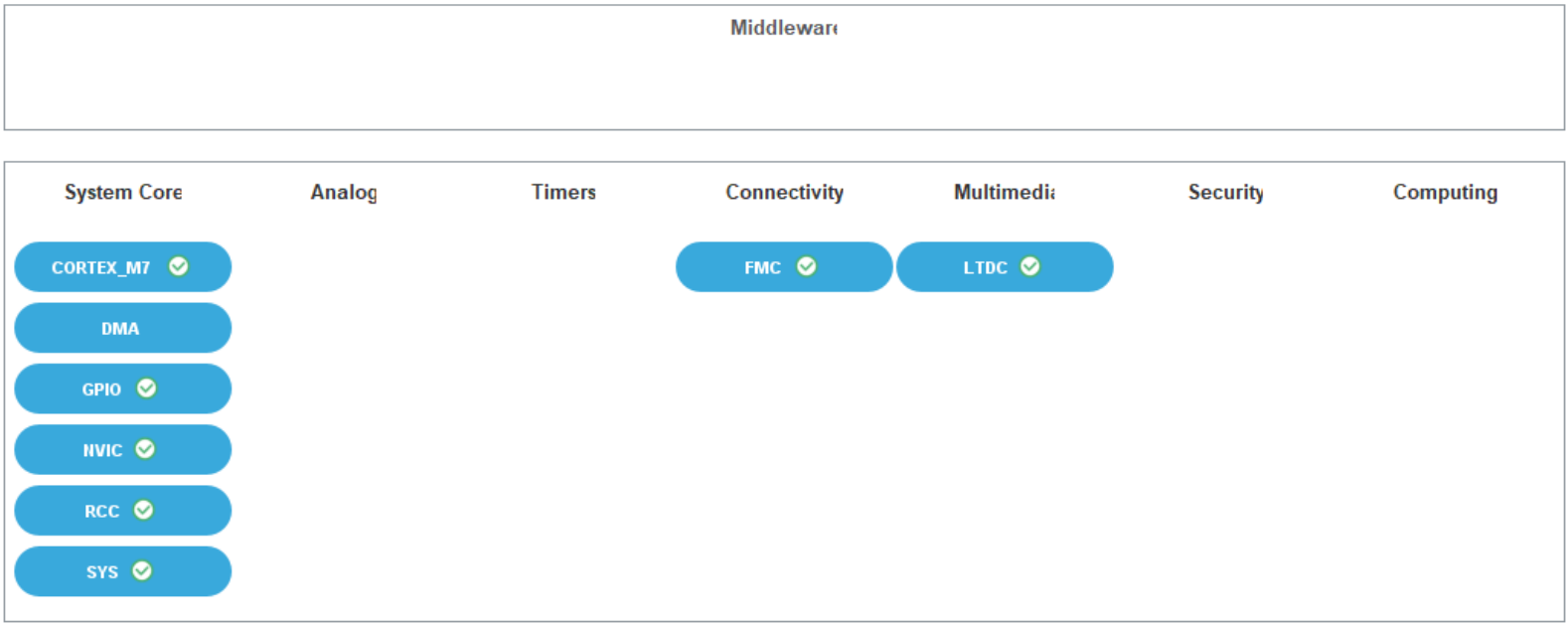
| Enabled interrupt Table | Select for init sequence ordering | Generate IRQ handler | Call HAL handler |
|---|---|---|---|
| Non maskable interrupt | false | true | false |
| Hard fault interrupt | false | true | false |
| Memory management fault | false | true | false |
| Pre-fetch fault, memory access fault | false | true | false |
| Undefined instruction or illegal state | false | true | false |
| System service call via SWI instruction | false | true | false |
| Debug monitor | false | true | false |
| Pendable request for system service | false | true | false |
| System tick timer | false | true | true |

*** User modified value**

# 5. System Views

5.1. Category view

*5.1.1. Current*



# 6. Docs & Resources

| Type | Link |
|---|---|
| BSDL files | https://www.st.com/resource/en/bsdl_model/stm32f7_bsdl.zip |
| IBIS models | https://www.st.com/resource/en/ibis_model/stm32f7_ibis.zip |
| System View Description | https://www.st.com/resource/en/svd/stm32f7-svd.zip |
| Presentations | https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf |
| Presentations | https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf |
| Presentations | https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf |
| Presentations | https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf |
| Presentations | https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf |
| Flyers | https://www.st.com/resource/en/flyer/flstm32nucleo.pdf |
| Flyers | https://www.st.com/resource/en/flyer/flstm32trust.pdf |
| Flyers | https://www.st.com/resource/en/flyer/flstm32gui.pdf |
| Security Bulletin | https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4031-using-the-stm32f2-stm32f4-and-stm32f7-series-dma-controller-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32- |

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|---|---|
| | applications-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4661-getting-started-with-stm32f7-series-mcu-hardware-development-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4676-stm32f7-series-peripheral-interconnections-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4731-stm32cube-mcu-package-examples-for-stm32f7-series-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4839-level-1-cache-on-stm32f7-series-and-stm32h7-series-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4850-stm32-mcus-spreadspectrum-clock-generation-principles-properties-and-implementation-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4946-migration-of-microcontroller-applications-between-stm32f72xxxf73xxx-and-stm32f74xxxf75xxx-microcontrollers-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an5073-receiving-spdif-audio-stream-with-the-stm32f4f7h7-series-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an5293-migration-guide-from-stm32f7-series-to-stmh74x75x-stm32h72x73x-and-stmh7a37bx-devices-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4838-introduction-to- |

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| | memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4879-introduction-to-usb-hardware-and-pcb-guidelines-using-stm32-mcus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an5537-how-to-use-adc-oversampling-techniques-to-improve-signaltonoise-ratio-on-stm32-mcus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4230-introduction-to-random-number-generation-validation-using-the-nist-statistical-test-suite-for-stm32-mcus-and-mpus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an2867-guidelines-for-oscillator-design-on-stm8afals-and-stm32-mcusmpus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4013-introduction-to-timers-for-stm32-mcus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4277-how-to-use-pwm-shutdown-for-motor-control-and-digital-power-conversion-on-stm32-mcus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4861-introduction-to-lcdtft-display-controller-ltdc-on-stm32-mcus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4908-getting-started-with-usart-automatic-baud-rater-detection-for-stm32-mcus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an5156-introduction-to-security-for-stm32-mcus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an5543-guidelines-for-enhanced-spi-communication-on-stm32-mcus-and-mpus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/cd00211314-how-to-optimize-the-adc-accuracy-in-the-stm32-mcus-stmicroelectronics.pdf |

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| Application Notes | https://www.st.com/resource/en/application_note/an2639-soldering-recommendations-and-package-information-for-leadfree-ecopack2-mcus-and-mpus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an5020-introduction-to-digital-camera-interface-dcmi-for-stm32-mcus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an3154-how-to-use-can-protocol-in-bootloader-on-stm32-mcus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4566-how-to-extend-the-dac-performance-on-stm32-mcus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an5647-the-goertzel-algorithm-to-compute-individual-terms-of-the-discrete-fourier-transform-dft-in-stm32-products-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an2606-introduction-to-system-memory-boot-mode-on-stm32-mcus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an5036-guidelines-for-thermal-management-on-stm32-mcus-and-mpus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an3126-how-to-use-dac-for-audio-and-waveform-generation-on-stm32-mcus-and-mpus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an3155-how-to-use-usart-protocol-in-bootloader-on-stm32-mcus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an3156-how-to-use-usb-dfu-protocol-in-bootloader-on-stm32-mcus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4286-how-to-use-spi-protocol-in-bootloader-on-stm32-mcus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4759-introduction-to-using-the-hardware-realtime-clock-rtc-and-the-tamper-management-unit-tamp-for-stm32-mcus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4760-introduction-to-quadspi-interface-for-stm32-mcus-and-mpus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4776-how-to-use-generalpurpose-timer-peripheral-on-stm32-mcus-stmicroelectronics.pdf |
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| Application Notes | https://www.st.com/resource/en/application_note/an4660-migrating-from-stm32f42xxx43xxx-to-stm32f7-mcus-stmicroelectronics.pdf |
| Errata Sheets | https://www.st.com/resource/en/errata_sheet/es0290-stm32f74xxx-and-stm32f75xxx-device-limitations-stmicroelectronics.pdf |
| Datasheet | https://www.st.com/resource/en/datasheet/dm00166116.pdf |
| Programming Manuals | https://www.st.com/resource/en/programming_manual/pm0253-stm32f7-series-and-stm32h7-series-cortexm7-processor-programming-manual-stmicroelectronics.pdf |
| Reference Manuals | https://www.st.com/resource/en/reference_manual/rm0385-stm32f75xxx-and-stm32f74xxx-advanced-armbased-32bit-mcus-stmicroelectronics.pdf |
| Technical Notes & Articles | https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf |
| Technical Notes & Articles | https://www.st.com/resource/en/technical_note/tn1205-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-fpn-packages-stmicroelectronics.pdf |
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| User Manuals | https://www.st.com/resource/en/user_manual/um2318-stm32f7-series-safety-manual-stmicroelectronics.pdf |
| User Manuals | https://www.st.com/resource/en/user_manual/um3251-stm32f7-series-ulcsaiec-607301603351-selftest-library-user-guide-stmicroelectronics.pdf |